18/09/2025

Types of Cross Validation

→ While training our model we split d. set into Train, test.

for the grouping we use an other named random state which makes sure of the perfect alignment division of T-T split such that accuracy of our model is increased

say, for 1000 records

at random state = 0 . 70% 30% T-T split } → accuracy ⇒ 85%

random state = 50 . 75% 25% TTS } accuracy ⇒ 87%

This fluctuation in the accuracy causes the model to get confused.

→ We change or stop fluctuations using Cross Validation techniques

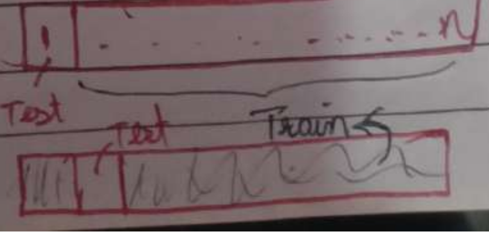
① Leave One Out Cross Validation (LOO CV).

Variance ↑
Bias ↓ .

say, we get 1000 records.

Exp.1}

Exp.2}



K=5

(11) K Fold CV

Test → Train

Exp 1 Accuracy → 1 | 200 | 800 | 1000.

Exp 2 Accuracy → 2 | 200 | 200 | 600 | - 1000

Acc 3

⋮

Acc K.

$$\left\{ \sum_{i=1}^{K} Acc\ i \Big/ K \right\}$$ ⇒ Mean of accuracies which would eventually tell us for a particular random test value, what's our accuracy.

*Disadvantages → Say if we get an imbalanced d-sets, we can not actually commit or summarise the mean as it would give a bad accuracy!!

PTO.

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(iii) Stratified Cross Validation

Say for 1000 records of d. set.

400 No's 600 Yes's.

S CV will allow equal & ~~sy~~ valid proportions of Yes's & No's in both Test and train Sets

~~Train~~ Test. → Train → No. of instances of each class.

60-40 60-40.

For Yes - NO

(iv) Time - Series CV.

O/P

Day 1 [D1 D2 D3 D4 D5] D6

[D1 D2 D3 D4 D5 D6. D7

Day 6
Test Pred: Day 7